

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_1204.parquet

GRU Multi-Step Training Report

Report Generated: 2025-05-06 17:08:37

Training Start: 2025-05-06 16:54:40

Training End: 2025-05-06 17:05:17

Training Duration: 10.62 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.5113

MSE: 3.7596

RMSE: 1.9390

R2 Score: 0.7794

Multi-Step Forecast:

Forecast Steps: 20

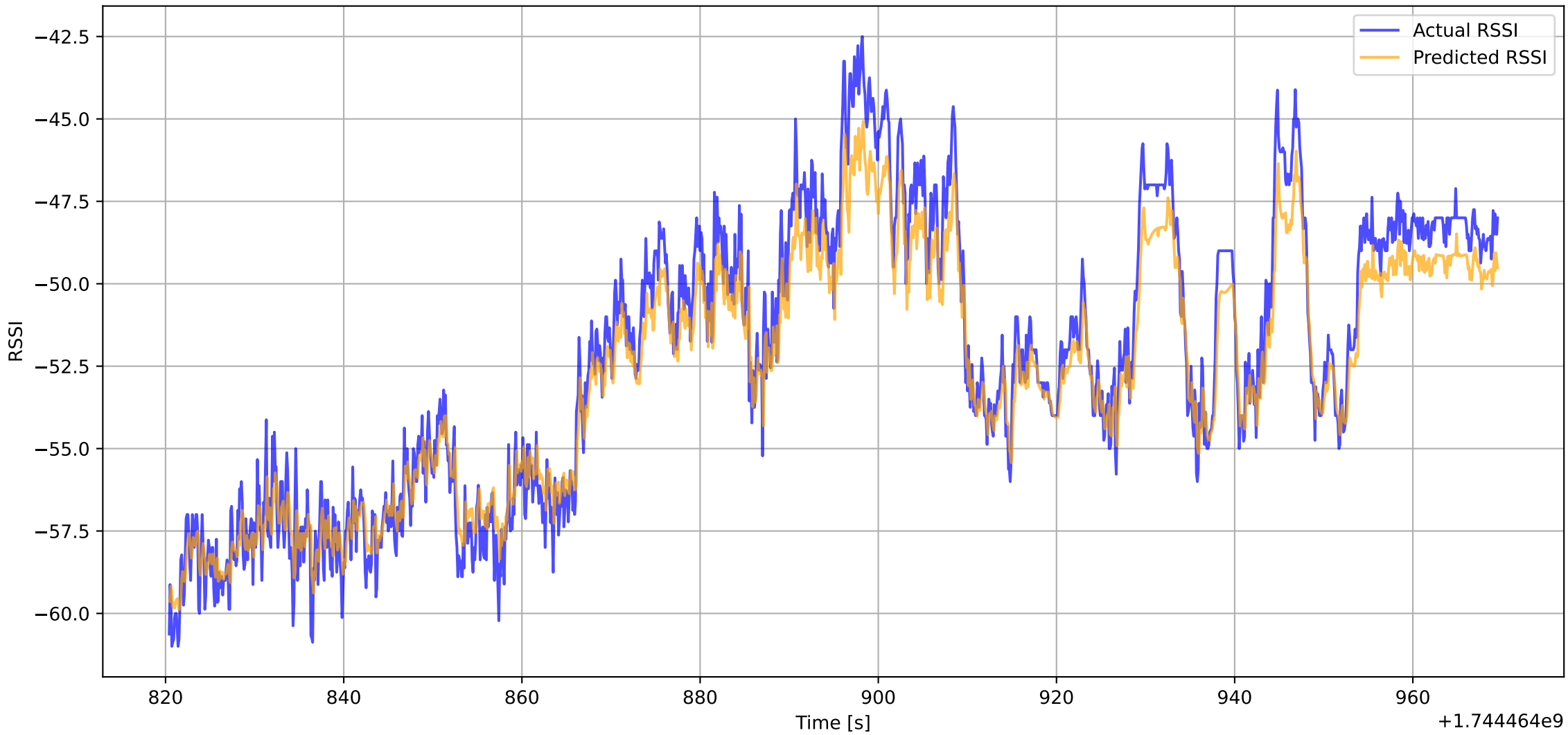
MAE: 0.0342

MSE: 0.0018

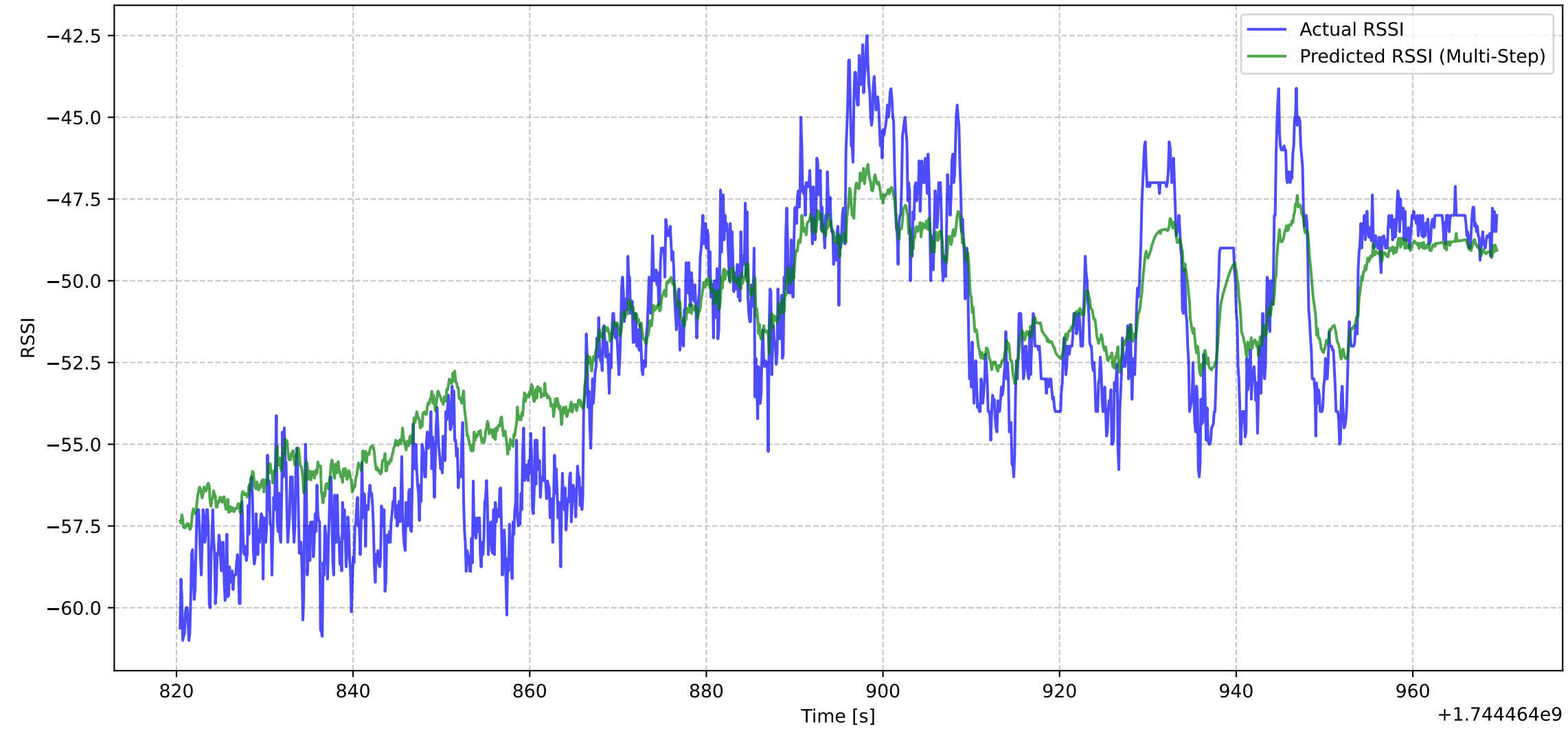
RMSE: 0.0425

R2 Score: 0.9233

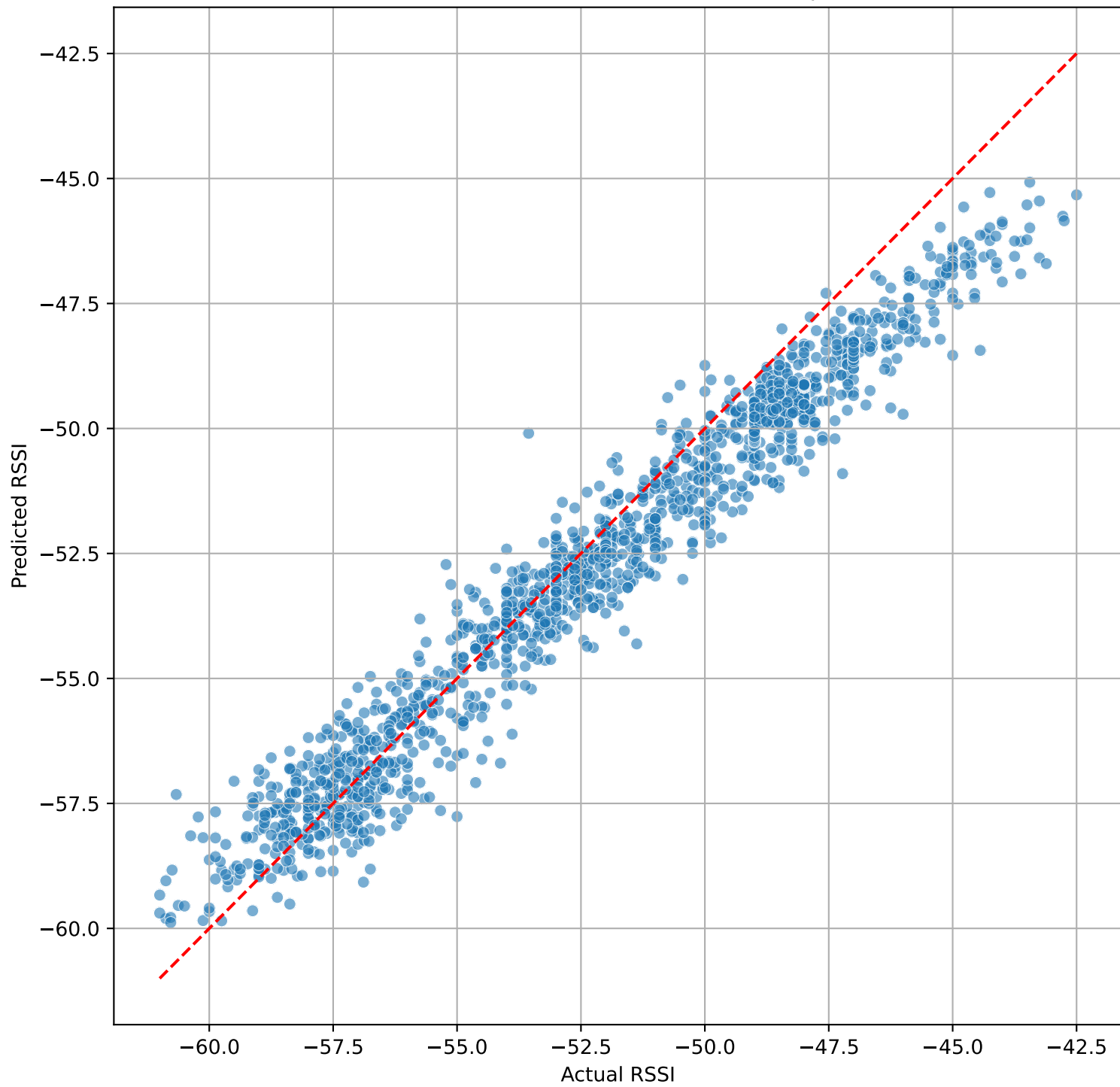
Actual vs Predicted RSSI (First Step)



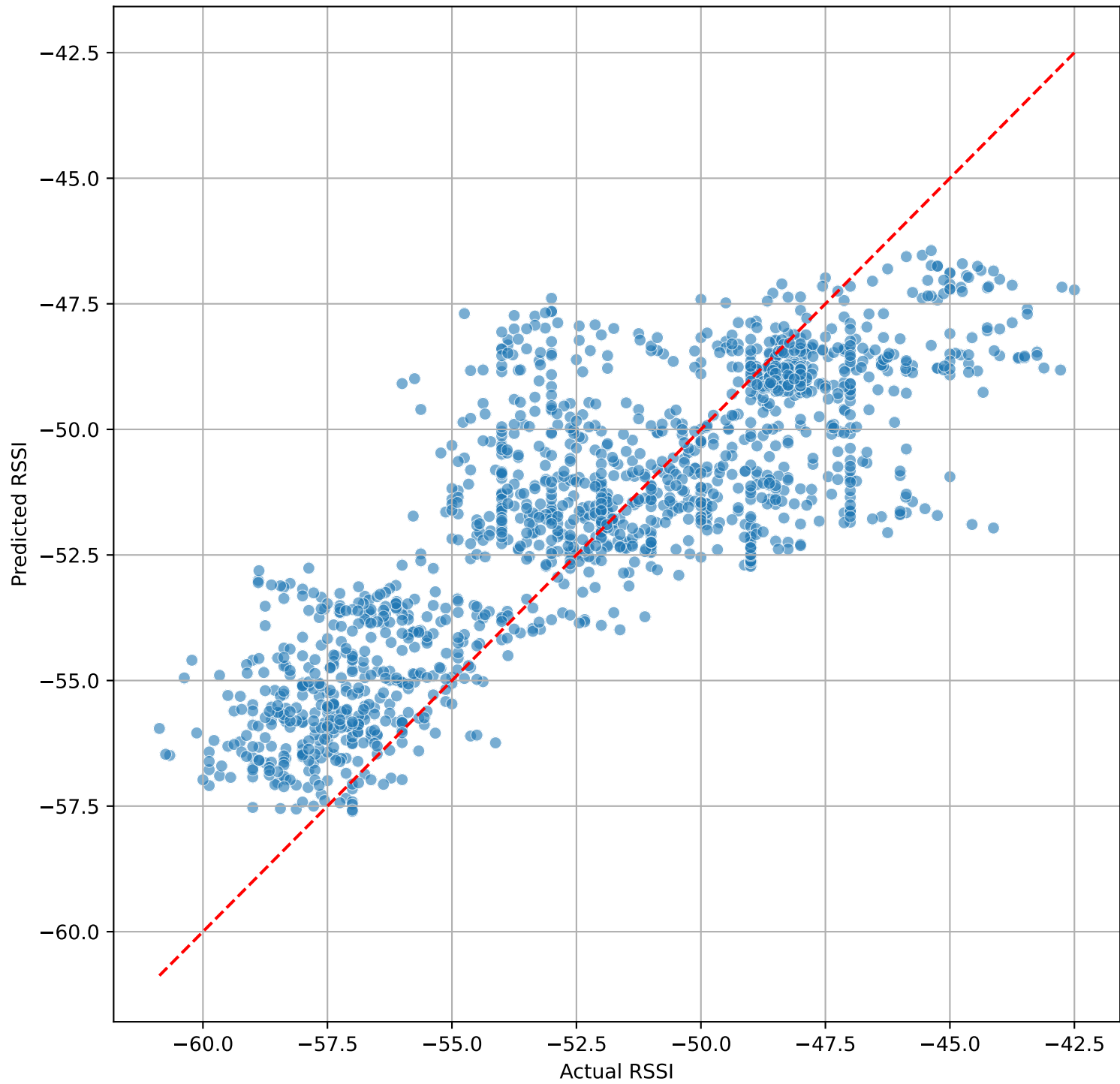
Actual vs Predicted RSSI (Multi-Step (step 20))



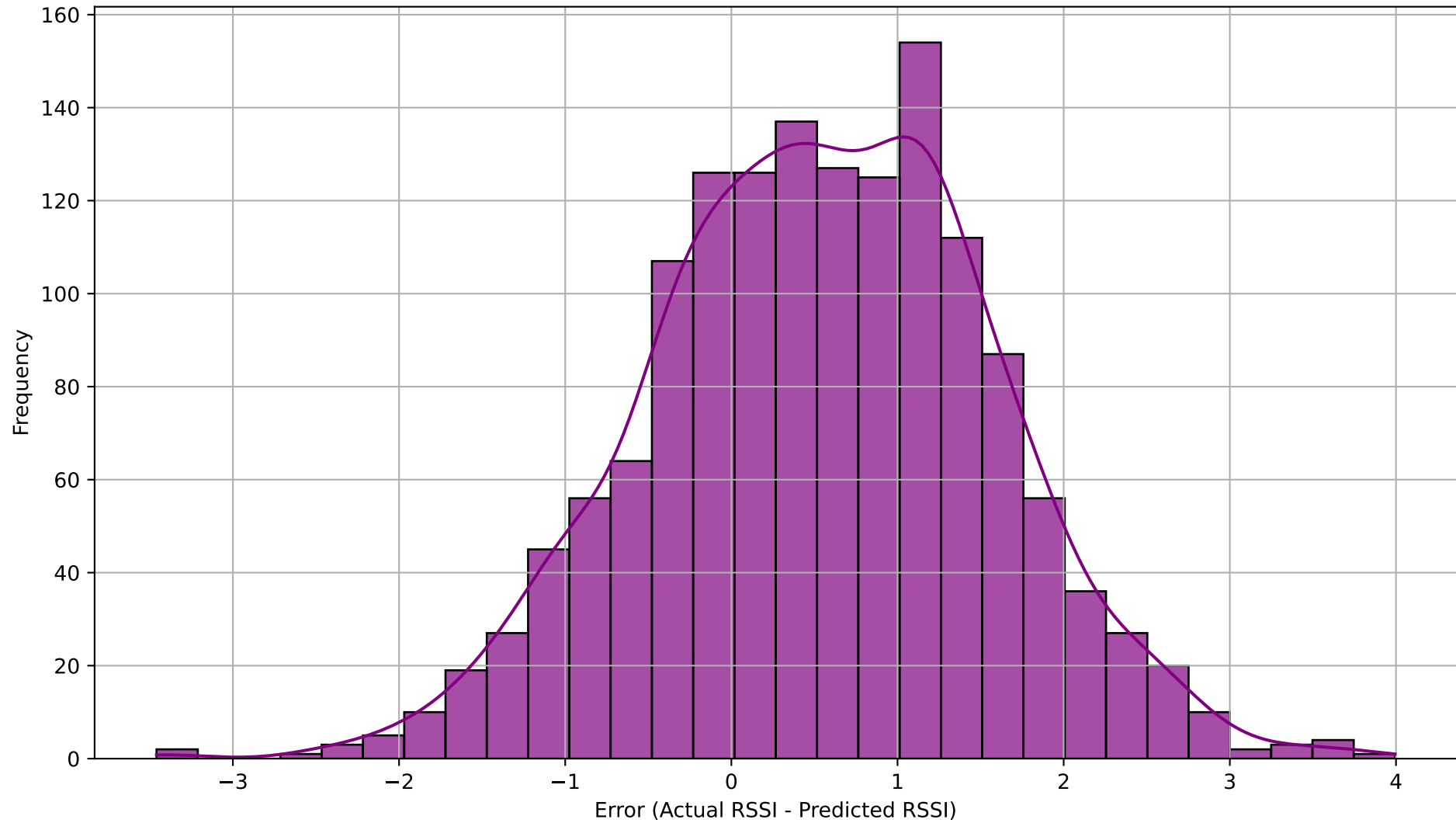
Actual vs Predicted RSSI (First Step)



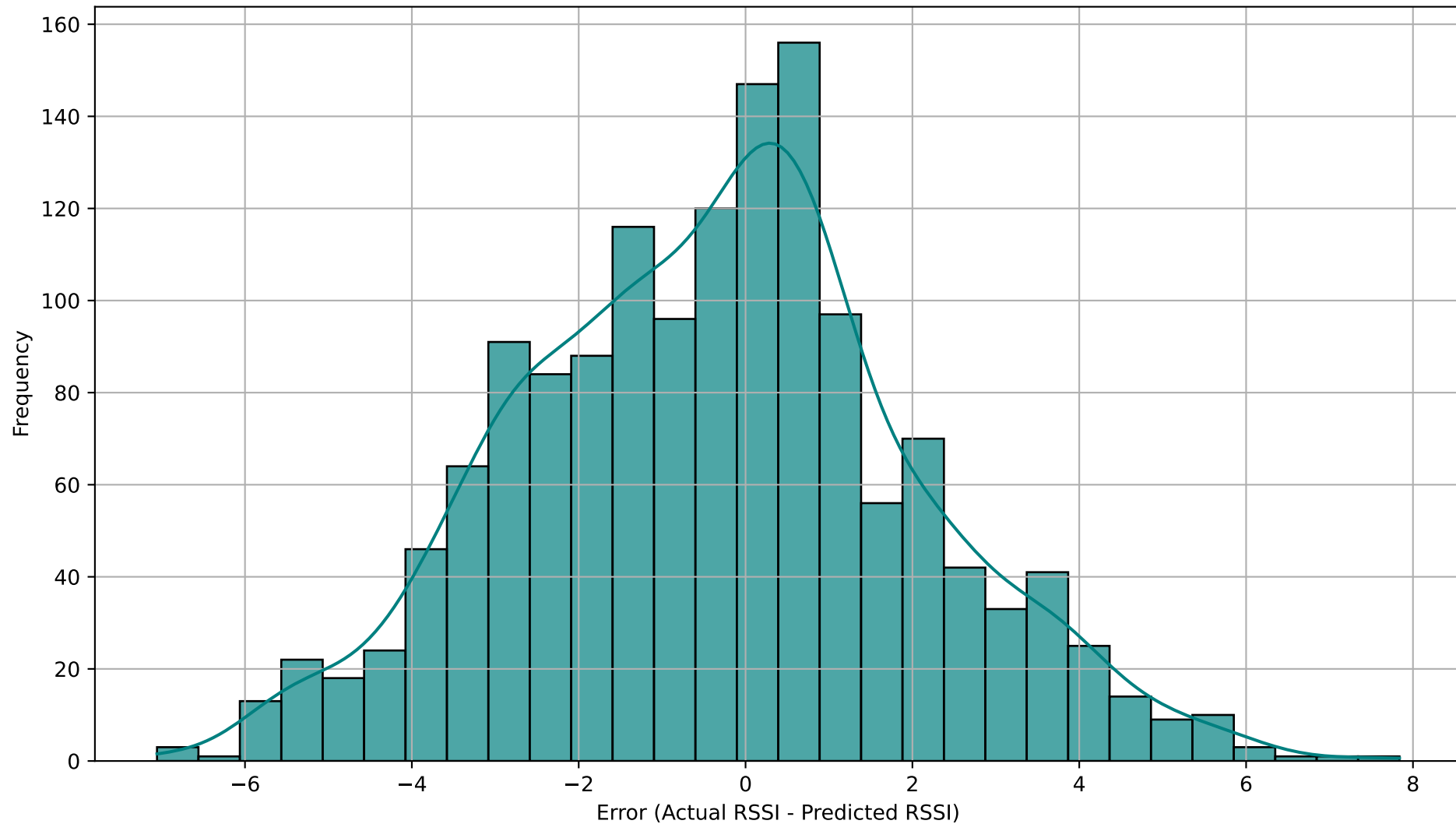
Actual vs Predicted RSSI (Last Step - step 20)



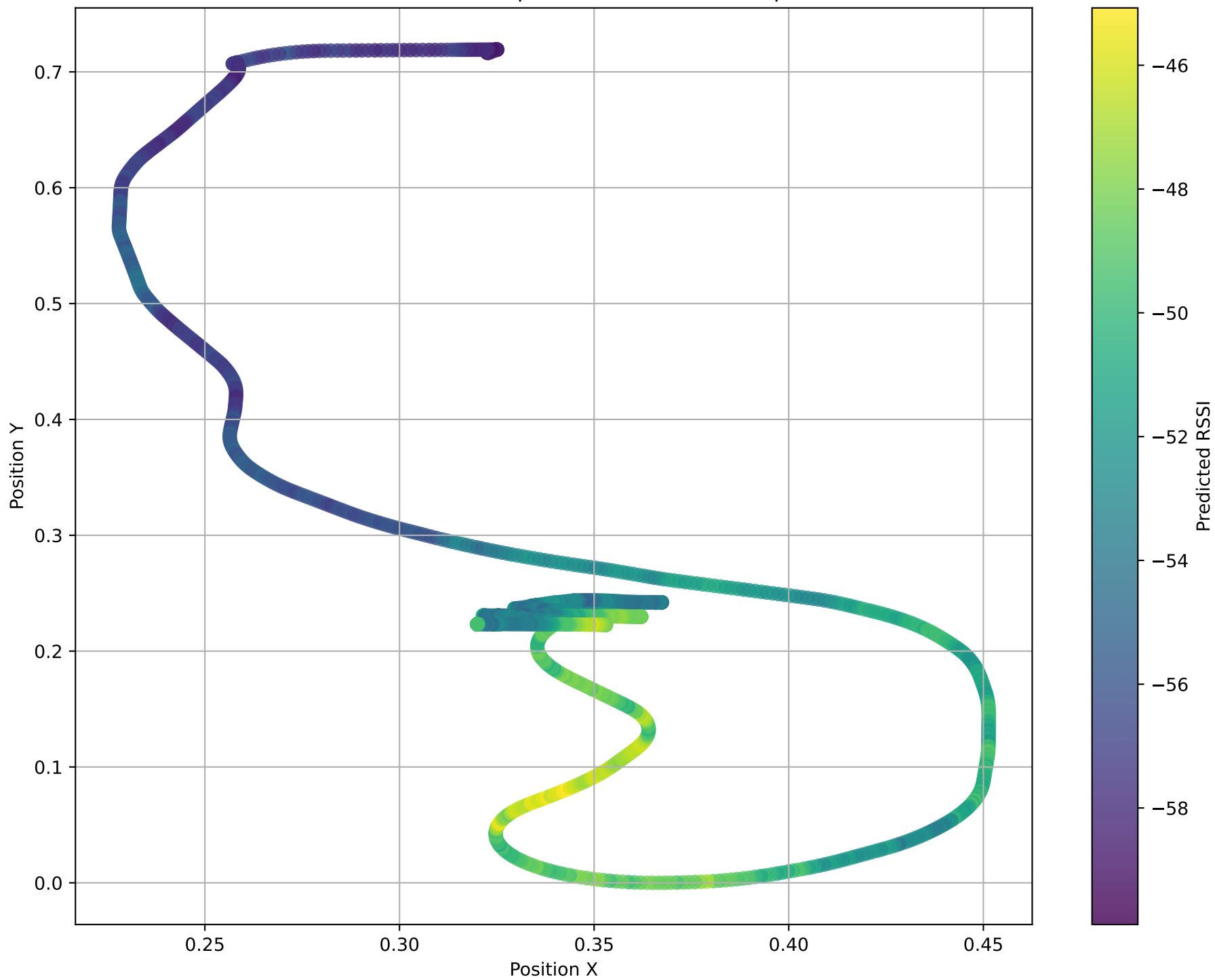
Distribution of Prediction Errors (First Step)



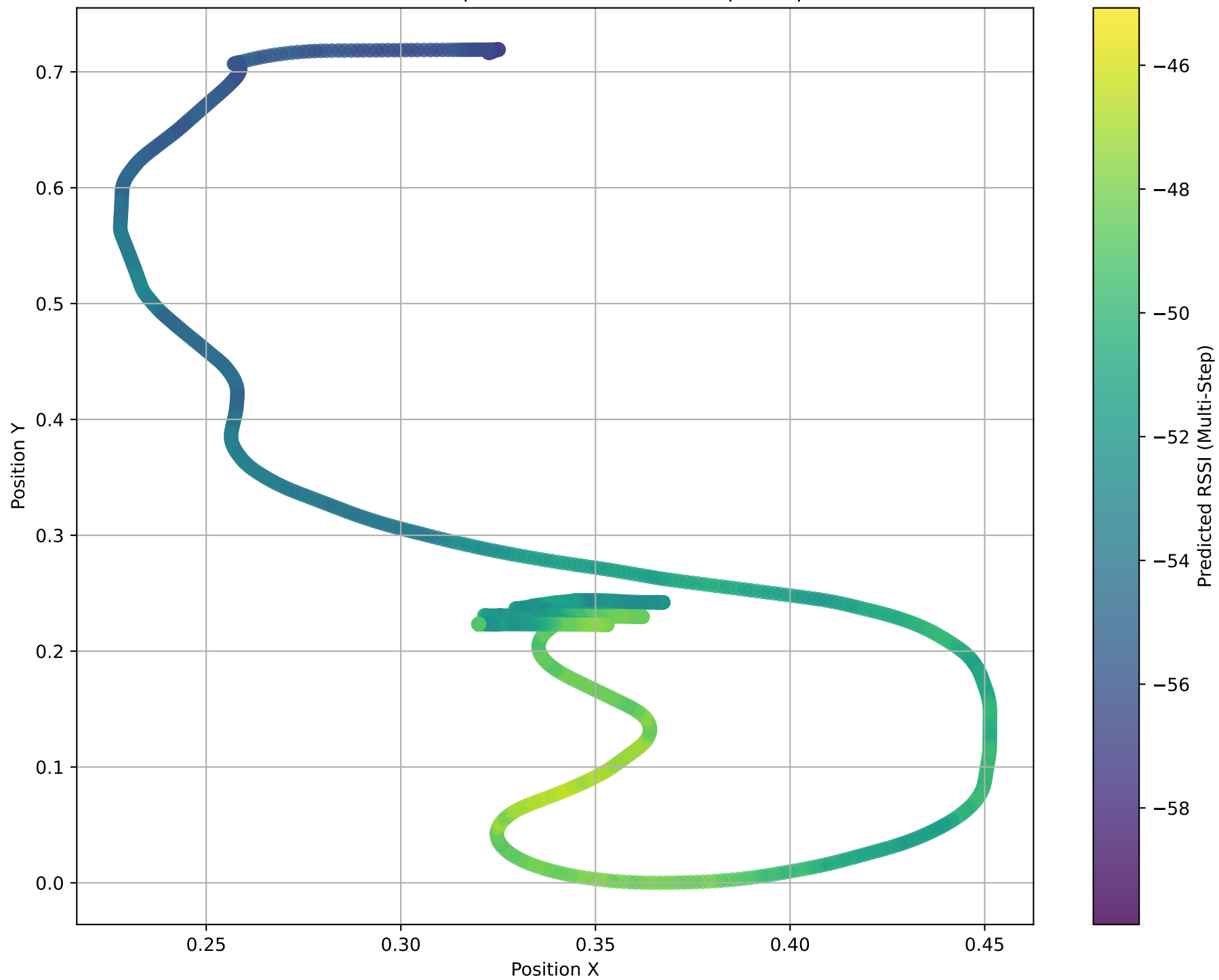
Distribution of Prediction Errors (Last Step - step 20)



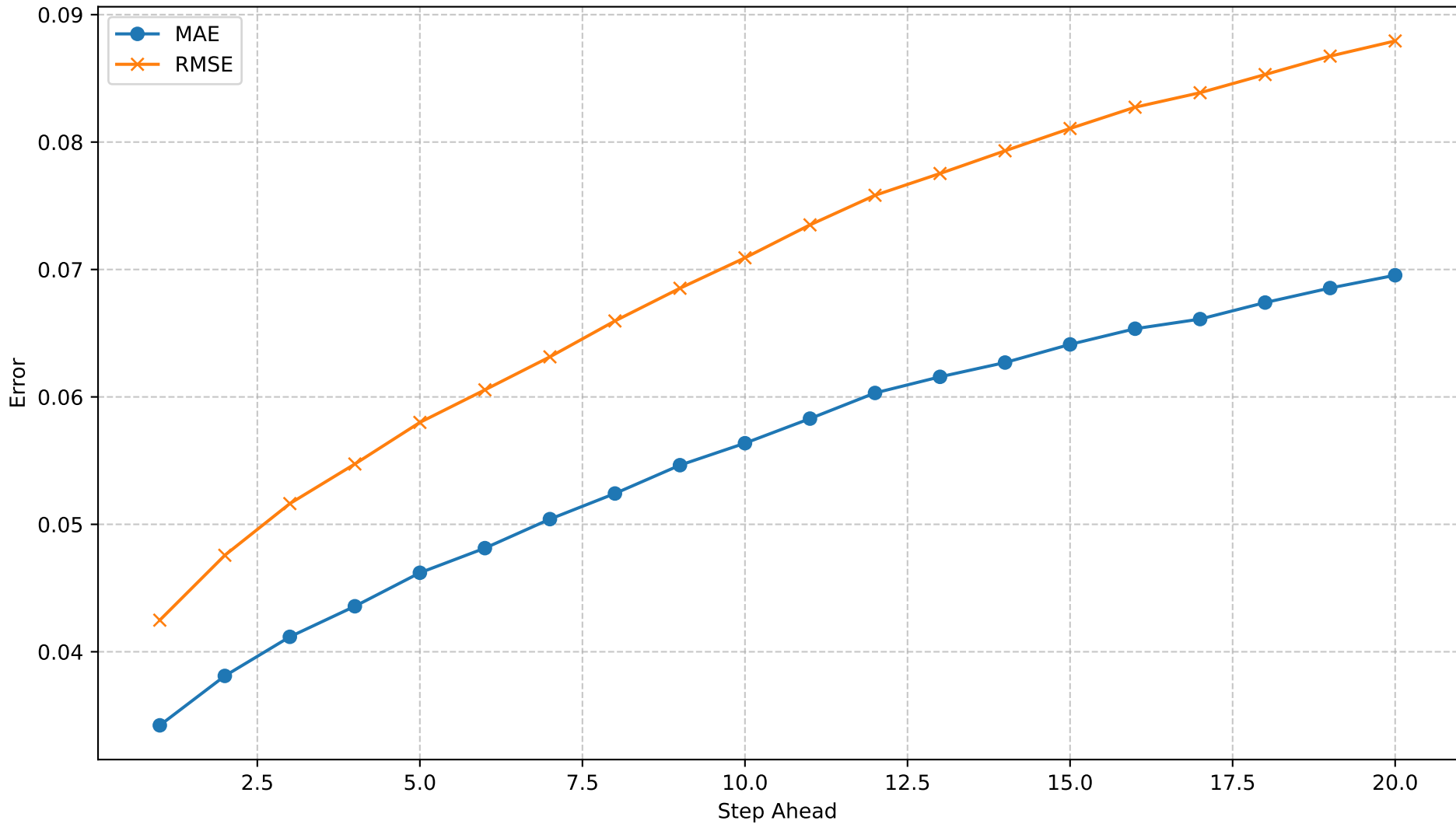
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_1204.parquet

Modelo: GRU

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.06 / 0.07
- 20º passo : 0.07 / 0.09

Crescimento do erro: 103.2% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.07), 19 (0.07), 18 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.